

Scottish FA UEFA Licence Tactical Platform

Official Platform Manual & Dynamic Scaffolding Guide
UEFA C • UEFA B • UEFA A Licence Tiers

Document Purpose: Complete operational guide for coach candidates, coach educators, and technical directors utilizing the Scottish FA UEFA Licence Tactical Platform. Outlines dynamic licence scaffolding, interactive pitch controls, frame-by-frame keyframe animations, TPPS player profiling, drill architectures, and automated opponent simulation engines.

Author / Directorate:	Scottish FA Coach Education & Technical Department
Supported Licence Tiers:	UEFA C Licence, UEFA B Licence, UEFA A Licence
Core Methodology:	The Coach, The Environment, The Player, The Game (4-Pillar Model)
Software Release:	Version 1.0.0 (Production Build - React 19 / Vite)
Live Deployment:	https://oliveannandale-max.github.io/scottish-fa-tactics/
Repository:	https://github.com/oliveannandale-max/scottish-fa-tactics

1. Executive Summary & Core Philosophy

The **Scottish FA UEFA Licence Tactical Platform** is a dynamic coaching application engineered to provide candidates with a high-fidelity tactical board while structurally enforcing the Scottish Football Association’s coach education syllabus. Rather than serving as an unconstrained graphic drawing tool, the software incorporates **Dynamic Scaffolding**—an intelligent rules engine that tailors available mechanics, tactical constraints, and analytical depths to the candidate’s enrolled licence tier (UEFA C, UEFA B, or UEFA A).

The Scottish FA 4-Pillar Coaching Model

Every exercise, tactical animation, and assessment sheet generated by this platform aligns with four fundamental pillars:

Pillar	Key Dimension	Platform Implementation
The Coach	Behaviours, communication, and intervention methodology.	Interactive coaching styles: <i>Drive-in</i> , <i>Freeze & Replay</i> , <i>Concurrent</i> , and <i>Terminal</i> .
The Environment	Culture, pitch geometry, psych climate, and learning constraints.	Dynamic pitch views: Full, Half, Penalty Box, Thirds, and Corridors.
The Player	Technical, Physical, Psychological, Social (TPPS) profiling.	Integrated TPPS 4-Corner radar analysis & individual developmental notes.
The Game	Principles of play, transitions, and phased tactical problem-solving.	Multi-phase session builder, opponent AI pressure modeling, and animation engine.

2. Dynamic Tier Scaffolding Matrix

The core innovation of the platform is that **UI complexity directly mirrors coaching competency expectations**. As a candidate transitions from UEFA C to UEFA A, UI constraints dynamically expand to accommodate higher-order tactical reasoning.

Feature / Constraint	UEFA C Licence	UEFA B Licence	UEFA A Licence
Target Domain	Fundamentals, 1v1 to 5v5 small practices, youth & community.	Units of play, 7v7 to 9v9, functional practices, phase play.	11v11 macro strategy, opponent game plans, periodisation.
Player Capacity	Max 10 outfielders (Small-Sided Games)	Max 16 players (Positional units & phased overloads)	Full 22 players + Substitutes & Goalkeeping units
Pitch Geometry	Full pitch, Half pitch, Penalty box zone.	Thirds of the pitch, Horizontal channels & corridors.	Complete tactical zones, 18-box grid & half-spaces.
Animation Engine	Single-step ball & player transitions.	Multi-frame sequence recording (up to 8 keyframes).	Continuous keyframe timeline, variable speeds & looping.
Tactical Markers	Solid pass line, simple run arrows.	Dashed runs, dribble squiggles, pressing zones.	Half-space shaded zones, offside lines, press triggers.
Opponent AI Presets	Static mannequins & passive opposition.	Reactive Mid-Block 4-4-2, High Press 4-3-3.	Deep Low-Block 5-4-1, Counter-Press 3-5-2, Custom Models.

3. Interactive Tactical Pitch Studio

The Interactive Pitch is rendered in high-definition SVG with hardware-accelerated drag-and-drop mechanics. It serves as the primary canvas for session geometry and drill visualization.

Pitch Views & Geometry Modes

Use the top-bar pitch selector to instantly reorient session boundaries:

View Mode	Visual Scope	Methodological Application
Full Pitch	105m x 68m standard pitch with touchline markings.	11v11 match simulations, transition sequences, counter-attack modeling.
Half Pitch	One half from halfway line to goal line.	Attacking against deep blocks, phased buildup, defensive unit organization.
Penalty Box / Final Third	Focused on 18-yard box, penalty arc, and flanks.	Finishing drills, crossing & cutback routines, defending crosses, set-pieces.
Thirds View	Horizontal dividing lines (Defending, Middle, Attacking).	Scottish FA unit linking, zone transition rules (e.g. 3-touch in middle third).
Channelled / Corridors	5 vertical lanes (Left Flank, Half-Space, Central, Right Half-Space, Right Flank).	Positional play, half-space penetration, underlapping fullbacks.

Player & Equipment Palette Controls

- **Adding Players:** Drag from the sidebar player dock onto the pitch. Players feature team color identifiers (Blue Home, Red Away, Yellow Neutral/Floater, Green Goalkeeper) and squad numbers.
- **Positioning & Moving:** Click and drag any player across pitch zones. Positions automatically calculate percentages relative to pitch dimensions for consistent rendering across screen resolutions.
- **Equipment Elements:** Place official training equipment including Standard Goals, Mini Pug Goals, Flat Discs/Cones (Orange/Yellow), Passing Mannequins, and Agility Hurdles.
- **Player Inspection:** Click on any player to open the contextual TPPS quick-inspector modal for tactical role assignment.

Vector Tactical Drawing Suite

Toggle the pencil tool to activate the vector drawing overlay with dedicated tactical stroke styles:

Tool	Visual Stroke	Tactical Meaning in SFA Syllabus
Pass Arrow	Solid crisp line with arrowhead	Ball trajectory, ground pass, or aerial delivery.
Movement Run	Dashed / perforated line with arrowhead	Off-the-ball player run (overlapping, blind-side, decoy).
Dribble Line	Wavy / undulating curve with arrowhead	Player carrying or driving into space with the ball.
Pressing Zone	Translucent shaded polygon overlay	Target defensive entrapment zone or high-intensity press area.
Tactical Area	Rectangular highlight with dashed boundary	Designated drill boundary, overload zone, or scoring channel.

4. Frame-by-Frame Keyframe Animation Engine

Static session diagrams often fail to convey speed of thought, anticipation, and secondary runs. The built-in **Keyframe Animation Studio** allows coaches to record multi-step tactical phases and replay them smoothly.

Step-by-Step Animation Workflow

Step 1: Set Initial Positions (Frame 1)

Position players and equipment on the pitch to represent the starting phase of the practice or game moment.

Step 2: Capture Keyframe

Click the '+ **Capture Frame**' button. The timeline stores the snapshot as Frame 1.

Step 3: Move Elements (Frame 2)

Drag the ball to the receiver, move defending units into compact blocks, and advance attacking runners.

Step 4: Capture Subsequent Frames

Repeat capture across up to 8 keyframes (UEFA B) or unlimited frames (UEFA A).

Step 5: Playback & Verification

Press **Play** to observe smooth interpolated motion. Adjust playback speed between **0.5x** (detailed coaching review), **1.0x** (game speed), and **2.0x** (quick overview).

5. Session & Drill Design Studio

The Scottish FA mandates that every training practice adheres to structured pedagogical frameworks. The Session Builder enforces this rigor through standardized fields and validation checks.

Session Metadata & SFA Coaching Intervention Matrix

Intervention Style	Trigger Moment	Candidate Coaching Action
Freeze & Replay	Major tactical breakdown or optimal learning picture.	Blow whistle, freeze players in exact positions, ask guided discovery questions, replay sequence with correct option.
Drive-In / Coaching in the Game	Targeted player requires individual instruction without stopping game.	Coach steps onto pitch boundary, provides direct technical correction to individual player while drill continues.
Concurrent Coaching	Positive reinforcement or reminders during active play.	Short, punchy verbal cues ('Check shoulder', 'Body shape', 'Set') without breaking player concentration.
Terminal Intervention	Interval, water break, or session conclusion.	Gather squad, review key outcomes, facilitate peer reflection, check tactical understanding.

The SFA '5Ws' Tactical Framework

Every Scottish FA assessment requires candidates to explicitly define the 5Ws for their practice:

- **WHAT:** The specific technical or tactical problem (e.g. Failure to break lines through central midfield).
- **WHERE:** The precise pitch location where the problem occurs (e.g. Middle third into attacking third).
- **WHO:** The specific units, positions, and numbers involved (e.g. 6 & 8 vs opposing 10 & 9).
- **WHEN:** The match moment/trigger (e.g. Immediately upon transitioning from defending to attacking).
- **WHY:** The underlying tactical rationale (e.g. To create central overloads and destabilize opposing double pivot).

6. Player Profiling & TPPS Radar System

Modern UEFA coach education prioritizes player-centered development. The platform integrates a **TPPS 4-Corner Radar System** allowing coaches to evaluate outfielders and goalkeepers across the four pillars of player capability.

TPPS Dimension	Evaluated Competencies	SFA Coaching Focus
Technical (T)	Ball mastery, 1st touch directional control, range of passing, finishing.	Execution under cognitive pressure and variable spatial constraints.
Physical (P)	Agility, acceleration, deceleration, core stability, aerial power, stamina.	Tactical periodisation load, work-to-rest ratios per drill.
Psychological (P)	Decision-making speed, emotional regulation, resilience under adversity.	Creating learning climates that embrace productive failure and problem-solving.
Social (S)	Communication, leadership, cohesion within units, coachability.	Peer mentoring, verbal/non-verbal signaling, collective responsibility.

7. Opponent Simulation & AI Tactical Engine

A hallmark of UEFA A and B Licence coaching is designing practices with realistic opposition behaviors. The Opponent AI engine simulates tactical shape and collective pressing shifts:

- **High Press 4-3-3:** Opposition wingers press inside-out forcing play into central traps; high defensive line compresses space.
- **Mid Block 4-4-2:** Two compact banks of four denying space between the lines; forwards drop to screen central defensive midfielders.
- **Low Block 5-4-1:** Dense penalty box protection, narrow defensive shape, and explosive counter-attacks into channels.
- **Counter-Attacking 3-5-2:** High wingback positioning, aggressive transitional overloads upon ball recovery.

8. GitHub Deployment, Sharing & Offline Access

The platform is distributed as a cloud-ready web application with automated GitHub Actions continuous integration:

Channel	URL / Command	Functionality
Live Cloud App	https://oliveannandale-max.github.io/scottish-fa-tactics/	Instant browser access on tablet, desktop, or touch displays.
Source Code	https://github.com/oliveannandale-max/scottish-fa-tactics	Complete React 19 / TypeScript / Vite source code and commit history.
Local Execution	git clone • npm install • npm run dev	Local high-speed development server at http://localhost:5173/ .

9. Candidate Quick Reference & Shortcuts

Key Combination	Action / Studio Function
Spacebar	Play / Pause tactical animation timeline
Delete / Backspace	Remove selected player, equipment item, or tactical drawing line

Ctrl + Z / Cmd + Z	Undo previous tactical placement or drawing stroke
Number Keys 1 - 3	Instantly switch Licence Tier (1: UEFA C, 2: UEFA B, 3: UEFA A)
P	Toggle Drawing Pencil Tool on/off
Esc	Close open modal (TPPS Radar, Mentor AI, Session Inspector)